

# Propositional Logic

*Notes By Pr. El Hadiq Zouhair*

## 2.1 Propositions

**Definition 2.1** (Proposition). A *proposition* is a declarative sentence that is either true (T) or false (F), but never both and never neither. Propositions are denoted by lowercase letters  $p, q, r, \dots$  and take their values in the set  $\mathbb{B} = \{\text{T}, \text{F}\}$  of *truth values*.

**Example 2.2.** “ $2 + 2 = 4$ ” is a true proposition; “Paris is the capital of Spain” is a false proposition. By contrast, “Close the door” (imperative), “What time is it?” (interrogative) and “ $x + 1 = 0$ ” (free variable — a *predicate*, Chapter 4) are not propositions. The self-referential sentence “This sentence is false” is also excluded: it cannot consistently receive either truth value, violating the principle of bivalence.

**Definition 2.3** (Atomic and compound). An *atomic* proposition cannot be decomposed using logical connectives; it is symbolised by a single letter. A *compound* proposition is built from atomic propositions using connectives such as  $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$ .

Propositional logic studies only how the truth of a compound proposition depends on its atoms; looking *inside* an atom is the job of predicate logic (Chapter 4).

### 2.1.1 Well-formed formulas

The propositional language can be defined by a grammar: every atomic letter is a well-formed formula (wff); if  $\varphi$  and  $\psi$  are wffs, so are  $\neg\varphi$ ,  $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ ,  $(\varphi \rightarrow \psi)$  and  $(\varphi \leftrightarrow \psi)$ ; nothing else is a wff. This inductive definition (a recursive set in the sense of Chapter 14) makes “formula” a precise mathematical object.

**Definition 2.4** (Truth assignment). A *truth assignment* (or *valuation*) is a function  $v : \{p, q, r, \dots\} \rightarrow \mathbb{B}$  assigning a truth value to every atomic proposition. A valuation extends uniquely to all wffs via the truth tables of the connectives.

## 2.2 The logical connectives

**Definition 2.5** (Negation). The *negation*  $\neg p$  (“not  $p$ ”) is true exactly when  $p$  is false.

**Definition 2.6** (Conjunction and disjunction). The *conjunction*  $p \wedge q$  (“ $p$  and  $q$ ”) is true exactly when both  $p$  and  $q$  are true. The *disjunction*  $p \vee q$  (“ $p$  or  $q$ ”) is true exactly when at least one of  $p, q$  is true (inclusive or).

**Definition 2.7** (Exclusive or). The *exclusive or*  $p \oplus q$  is true exactly when  $p$  and  $q$  have different truth values.

**Definition 2.8** (Conditional). The *conditional* (material implication)  $p \rightarrow q$  (“if  $p$  then  $q$ ”) is false only when  $p$  is true and  $q$  is false; in all other cases it is true.  $p$  is the *hypothesis* (antecedent),  $q$  the *conclusion* (consequent).

A conditional with a false hypothesis is vacuously true: “if  $1 = 2$  then I am the Pope” is a true proposition. The conditional asserts nothing about causality — only that the case ( $p$  true,  $q$  false) does not occur.

**Definition 2.9** (Biconditional). The *biconditional*  $p \leftrightarrow q$  (“ $p$  if and only if  $q$ ”) is true exactly when  $p$  and  $q$  have the same truth value.

**Definition 2.10** (NAND and NOR).  $p \uparrow q \equiv \neg(p \wedge q)$  (NAND, Sheffer stroke) and  $p \downarrow q \equiv \neg(p \vee q)$  (NOR, Peirce arrow).

The complete behaviour of the binary connectives:

| $p$ | $q$ | $p \wedge q$ | $p \vee q$ | $p \oplus q$ | $p \rightarrow q$ | $p \leftrightarrow q$ | $p \uparrow q$ |
|-----|-----|--------------|------------|--------------|-------------------|-----------------------|----------------|
| T   | T   | T            | T          | F            | T                 | T                     | F              |
| T   | F   | F            | T          | T            | F                 | F                     | T              |
| F   | T   | F            | T          | T            | T                 | F                     | T              |
| F   | F   | F            | F          | F            | T                 | T                     | T              |

## 2.3 Truth tables

To build the truth table of a compound proposition with  $n$  atomic letters, list all  $2^n$  valuations (conventionally from  $T \cdots T$  down to  $F \cdots F$ ), then evaluate subformulas from the innermost outward.

**Example 2.11.** Truth table of  $(p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$ :

| $p$ | $q$ | $p \rightarrow q$ | $\neg p$ | $\neg p \rightarrow \neg q$ | whole formula |
|-----|-----|-------------------|----------|-----------------------------|---------------|
| T   | T   | T                 | F        | T                           | T             |
| T   | F   | F                 | F        | T                           | F             |
| F   | T   | T                 | T        | F                           | F             |
| F   | F   | T                 | T        | T                           | T             |

The formula is true exactly when  $p$  and  $q$  agree — it is equivalent to  $p \leftrightarrow q$ .

### 2.3.1 Precedence

To limit parentheses, the standard precedence (strongest first) is:  $\neg$ , then  $\wedge$ , then  $\vee$ , then  $\rightarrow$ , then  $\leftrightarrow$ . Thus  $\neg p \vee q \wedge r \rightarrow s$  reads  $((\neg p) \vee (q \wedge r)) \rightarrow s$ . The conditional associates to the right:  $p \rightarrow q \rightarrow r$  means  $p \rightarrow (q \rightarrow r)$ . When in doubt, parenthesize.

## 2.4 Converse, inverse, contrapositive

**Definition 2.12.** For the conditional  $p \rightarrow q$ : the *converse* is  $q \rightarrow p$ ; the *inverse* is  $\neg p \rightarrow \neg q$ ; the *contrapositive* is  $\neg q \rightarrow \neg p$ .

**Theorem 2.13.** A conditional is logically equivalent to its contrapositive:  $p \rightarrow q \equiv \neg q \rightarrow \neg p$ . It is *not* in general equivalent to its converse or to its inverse (which are equivalent to each other).

Two classical fallacies arise from confusing these forms: *affirming the consequent* (from  $p \rightarrow q$  and  $q$ , wrongly concluding  $p$ ) and *denying the antecedent* (from  $p \rightarrow q$  and  $\neg p$ , wrongly concluding  $\neg q$ ).

## 2.5 Tautology, contradiction, contingency

**Definition 2.14.** A compound proposition is a *tautology* if it is true under every valuation; a *contradiction* if it is false under every valuation; a *contingency* if it is neither.

**Example 2.15.**  $p \vee \neg p$  is a tautology (law of excluded middle);  $p \wedge \neg p$  is a contradiction;  $p \rightarrow q$  is a contingency.

**Definition 2.16** (Satisfiability). A formula is *satisfiable* if at least one valuation makes it true. The *SAT problem* asks whether a given formula is satisfiable; it is the canonical NP-complete problem, and a formula with  $n$  variables has  $2^n$  candidate valuations — brute force quickly becomes infeasible.

## 2.6 Logical equivalence

**Definition 2.17.** Two formulas  $\varphi$  and  $\psi$  are *logically equivalent*, written  $\varphi \equiv \psi$ , if they have the same truth value under every valuation — equivalently, if  $\varphi \leftrightarrow \psi$  is a tautology.

**Theorem 2.18** (Major equivalence laws). For all formulas  $p, q, r$  (with T, F denoting a tautology and a contradiction):

|                 |                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------|
| Identity        | $p \wedge \mathbf{T} \equiv p, \quad p \vee \mathbf{F} \equiv p$                               |
| Domination      | $p \vee \mathbf{T} \equiv \mathbf{T}, \quad p \wedge \mathbf{F} \equiv \mathbf{F}$             |
| Idempotence     | $p \vee p \equiv p, \quad p \wedge p \equiv p$                                                 |
| Double negation | $\neg(\neg p) \equiv p$                                                                        |
| Commutativity   | $p \vee q \equiv q \vee p, \quad p \wedge q \equiv q \wedge p$                                 |
| Associativity   | $(p \vee q) \vee r \equiv p \vee (q \vee r)$ , and dually for $\wedge$                         |
| Distributivity  | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ , and dually                         |
| De Morgan       | $\neg(p \wedge q) \equiv \neg p \vee \neg q, \quad \neg(p \vee q) \equiv \neg p \wedge \neg q$ |
| Absorption      | $p \vee (p \wedge q) \equiv p, \quad p \wedge (p \vee q) \equiv p$                             |
| Negation        | $p \vee \neg p \equiv \mathbf{T}, \quad p \wedge \neg p \equiv \mathbf{F}$                     |

**Theorem 2.19** (Implication and biconditional identities).

$$p \rightarrow q \equiv \neg p \vee q, \quad \neg(p \rightarrow q) \equiv p \wedge \neg q, \quad p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p).$$

Each law is proved by a truth table; once proved, laws may be chained to simplify formulas algebraically, replacing equals by equals.

## 2.7 Logical consequence

**Definition 2.20** (Entailment). A set of formulas  $\Gamma$  *entails*  $\varphi$ , written  $\Gamma \models \varphi$ , if every valuation making all formulas of  $\Gamma$  true also makes  $\varphi$  true. An *argument* with premises  $\Gamma$  and conclusion  $\varphi$  is *valid* exactly when  $\Gamma \models \varphi$ .

Validity can always be checked by a truth table, but with many atoms this is exponential; Chapter 5 develops *rules of inference* as a scalable alternative. The most important rules — modus ponens (from  $p$  and  $p \rightarrow q$ , infer  $q$ ), modus tollens (from  $\neg q$  and  $p \rightarrow q$ , infer  $\neg p$ ), hypothetical syllogism, disjunctive syllogism, addition, simplification, conjunction and resolution — are all entailments and are treated in detail there.

## 2.8 Normal forms

**Definition 2.21.** A *literal* is an atom or a negated atom. A formula is in *negation normal form* (NNF) if negations occur only on atoms. It is in *disjunctive normal form* (DNF) if it is a disjunction of conjunctions of literals, and in *conjunctive normal form* (CNF) if it is a conjunction of disjunctions of literals.

**Theorem 2.22.** Every propositional formula is equivalent to one in CNF and to one in DNF.

**Example 2.23** (CNF conversion). Convert  $\neg(p \rightarrow q) \vee r$  to CNF:

$$\neg(p \rightarrow q) \vee r \equiv (p \wedge \neg q) \vee r \equiv (p \vee r) \wedge (\neg q \vee r),$$

using Theorem 2.19 and then distributivity. The general algorithm: eliminate  $\leftrightarrow$  and  $\rightarrow$ , push negations inward by De Morgan (reaching NNF), then distribute  $\vee$  over  $\wedge$ .

A DNF can also be read directly off a truth table: each row where the formula is true contributes one conjunction of literals describing that row.

## 2.9 Functional completeness

**Definition 2.24.** A set of connectives is *functionally complete* if every Boolean function  $\mathbb{B}^n \rightarrow \mathbb{B}$  can be expressed using only connectives from the set.

**Theorem 2.25.**  $\{\neg, \wedge, \vee\}$  is functionally complete (via DNF); so are  $\{\neg, \wedge\}$  and  $\{\neg, \vee\}$  (by De Morgan); and remarkably each of the singletons  $\{\uparrow\}$  and  $\{\downarrow\}$  is functionally complete.

For NAND:  $\neg p \equiv p \uparrow p$  and  $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$ , which suffices. This is why entire processors can be built from NAND gates alone.

## 2.10 Logic gates and circuits

Each connective corresponds to a hardware *gate* (AND, OR, NOT, XOR, NAND, NOR). Compound formulas correspond to combinational circuits. The *half-adder*, which adds two bits, illustrates the correspondence: the sum bit is  $s = p \oplus q$  and the carry bit is  $c = p \wedge q$ . Cascading half-adders yields full adders and ultimately arithmetic units — arithmetic reduced to logic.

## 2.11 Translating English to logic

Useful patterns, with  $p$  “it rains” and  $q$  “I take an umbrella”:

- “If it rains, I take an umbrella”:  $p \rightarrow q$ .
- “I take an umbrella only if it rains”:  $q \rightarrow p$ .
- “I take an umbrella unless it rains”:  $\neg p \rightarrow q$ .
- “Taking an umbrella is necessary for rain”:  $p \rightarrow q$  ( $q$  is the necessary condition).
- “Taking an umbrella is sufficient for rain”:  $q \rightarrow p$ .
- “I take an umbrella if and only if it rains”:  $q \leftrightarrow p$ .

“Only if” introduces the conclusion, not the hypothesis; “unless” means “if not”. These two patterns cause most translation errors.

## 2.12 Summary

Propositions take values in  $\mathbb{B} = \{\mathbf{T}, \mathbf{F}\}$ ; connectives  $\neg, \wedge, \vee, \oplus, \rightarrow, \leftrightarrow, \uparrow, \downarrow$  combine them. Truth tables decide every semantic question in principle; equivalence laws (Theorem 2.18) decide them algebraically. Every formula has a CNF and a DNF;  $\{\uparrow\}$  alone suffices to express everything; and logic realised in silicon becomes digital circuitry.

## Exercises

**Exercise 2.1** EASY

Which of the following are propositions? (a) “7 is prime.” (b) “ $x^2 \geq 0$ .” (c) “Read this chapter twice.” (d) “ $1 + 1 = 3$ .” (e) “This statement is false.”

**Exercise 2.2** EASY

Let  $p$ : “the file compiles” and  $q$ : “the tests pass”. Express in symbols: (a) the file compiles but the tests fail; (b) if the file does not compile then the tests fail; (c) the tests pass only if the file compiles; (d) the file compiles if and only if the tests pass.

**Exercise 2.3** EASY

Build the truth table of  $(p \vee q) \wedge \neg(p \wedge q)$  and identify which standard connective it represents.

**Exercise 2.4** MEDIUM

State the converse, inverse and contrapositive of: “If  $n$  is divisible by 6, then  $n$  is divisible by 3.” Which of the four statements are true for all integers  $n$ ?

**Exercise 2.5** MEDIUM

Classify each formula as tautology, contradiction or contingency: (a)  $(p \rightarrow q) \vee (q \rightarrow p)$ ; (b)  $(p \wedge \neg p) \rightarrow q$ ; (c)  $(p \rightarrow q) \rightarrow p$ .

**Exercise 2.6** MEDIUM

Using the equivalence laws (not truth tables), simplify  $\neg(p \vee \neg q) \vee (\neg p \wedge \neg q)$  as far as possible, citing each law used.

**Exercise 2.7** MEDIUM

Prove using equivalence laws that  $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$ .

**Exercise 2.8** MEDIUM

Convert  $(p \wedge q) \rightarrow r$  to CNF, then to DNF, showing each step.

**Exercise 2.9** HARD

Show that  $\{\rightarrow, F\}$  is functionally complete, where  $F$  is the constant false. *Hint: first express  $\neg p$ , then  $\vee$ , then  $\wedge$ .*

**Exercise 2.10** HARD

Express  $p \vee q$  and  $p \rightarrow q$  using only the NOR connective  $\downarrow$ .

**Exercise 2.11** HARD

A formula has the truth table that is true exactly on the rows  $(p, q, r) = (T, T, F)$  and  $(F, T, T)$ . Write its DNF, and give an equivalent CNF.

**Exercise 2.12** HARD

Three suspects make statements. Alice: “Bob is lying.” Bob: “Carol is lying.” Carol: “Alice and Bob are both lying.” Using propositional variables  $a, b, c$  for “Alice (resp. Bob, Carol) tells the truth”, formalise the situation and determine who is telling the truth.

## Solutions to the Exercises

**Solution 2.1.** (a) Yes (true). (b) No as stated —  $x$  is a free variable, so it is a predicate; (it becomes a proposition once quantified, e.g.  $\forall x \in \mathbb{R}, x^2 \geq 0$ ). (c) No — imperative. (d) Yes (false). (e) No — it can receive no consistent truth value. ■

**Solution 2.2.** (a)  $p \wedge \neg q$ . (b)  $\neg p \rightarrow \neg q$ . (c)  $q \rightarrow p$ . (d)  $p \leftrightarrow q$ . ■

**Solution 2.3.**

| $p$ | $q$ | $p \vee q$ | $\neg(p \wedge q)$ | formula |
|-----|-----|------------|--------------------|---------|
| T   | T   | T          | F                  | F       |
| T   | F   | T          | T                  | T       |
| F   | T   | T          | T                  | T       |
| F   | F   | F          | T                  | F       |

True exactly when  $p$  and  $q$  differ: this is the exclusive or  $p \oplus q$ . ■

**Solution 2.4.** Converse: “if  $n$  is divisible by 3 then  $n$  is divisible by 6” — false ( $n = 9$ ). Inverse: “if  $n$  is not divisible by 6 then  $n$  is not divisible by 3” — false ( $n = 9$  again). Contrapositive: “if  $n$  is not divisible by 3 then  $n$  is not divisible by 6” — true, being equivalent to the original true statement. ■

**Solution 2.5.** (a) Tautology: if  $p \rightarrow q$  fails then  $p$  is true and  $q$  false, making  $q \rightarrow p$  true. (b) Tautology: the hypothesis  $p \wedge \neg p$  is always false, so the conditional is vacuously true. (c) Contingency: true when  $p = \text{T}$ ; false when  $p = \text{F}$  (then  $p \rightarrow q$  is true but the conclusion  $p$  is false). ■

**Solution 2.6.**

$$\begin{aligned}
 \neg(p \vee \neg q) \vee (\neg p \wedge \neg q) &\equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) && \text{De Morgan, double negation} \\
 &\equiv \neg p \wedge (q \vee \neg q) && \text{distributivity} \\
 &\equiv \neg p \wedge \text{T} && \text{negation law} \\
 &\equiv \neg p && \text{identity law.}
 \end{aligned}$$

**Solution 2.7.**

$$\begin{aligned}
 (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{implication identity} \\
 &\equiv (\neg p \wedge \neg q) \vee r && \text{distributivity} \\
 &\equiv \neg(p \vee q) \vee r && \text{De Morgan} \\
 &\equiv (p \vee q) \rightarrow r && \text{implication identity.}
 \end{aligned}$$

**Solution 2.8.**  $(p \wedge q) \rightarrow r \equiv \neg(p \wedge q) \vee r \equiv \neg p \vee \neg q \vee r$ . This single clause is simultaneously a CNF (one disjunctive clause) and a DNF (disjunction of three one-literal conjunctions). ■

**Solution 2.9.**  $\neg p \equiv p \rightarrow \mathbf{F}$ . Then  $p \vee q \equiv \neg p \rightarrow q \equiv (p \rightarrow \mathbf{F}) \rightarrow q$ , and  $p \wedge q \equiv \neg(\neg p \vee \neg q) \equiv (((p \rightarrow \mathbf{F}) \rightarrow (q \rightarrow \mathbf{F}))) \rightarrow \mathbf{F}$  after expanding  $\neg p \vee \neg q \equiv p \rightarrow \neg q$ , i.e.  $p \wedge q \equiv (p \rightarrow (q \rightarrow \mathbf{F})) \rightarrow \mathbf{F}$ . Since  $\{\neg, \wedge, \vee\}$  is functionally complete, so is  $\{\rightarrow, \mathbf{F}\}$ . ■

**Solution 2.10.**  $\neg x \equiv x \downarrow x$ . Then  $p \vee q \equiv \neg(p \downarrow q) \equiv (p \downarrow q) \downarrow (p \downarrow q)$ , and  $p \rightarrow q \equiv \neg p \vee q$ , so with  $A = p \downarrow p (= \neg p)$ :  $p \rightarrow q \equiv (A \downarrow q) \downarrow (A \downarrow q)$ . ■

**Solution 2.11.** DNF: each true row becomes a conjunction:  $(p \wedge q \wedge \neg r) \vee (\neg p \wedge q \wedge r)$ . For a CNF, note both rows have  $q = \mathbf{T}$ , and  $p \leftrightarrow \neg r$  holds on them ( $p$  and  $r$  differ); conversely  $q \wedge (p \oplus r)$  is true exactly on these two rows. Hence a CNF is  $q \wedge (p \vee r) \wedge (\neg p \vee \neg r)$ . ■

**Solution 2.12.** The statements give  $a \leftrightarrow \neg b$ ,  $b \leftrightarrow \neg c$ ,  $c \leftrightarrow (\neg a \wedge \neg b)$ . From the first two:  $a$  and  $b$  differ,  $b$  and  $c$  differ, so  $a = c$  and  $a \neq b$ . Case  $a = \mathbf{T}$ : then  $c = \mathbf{T}$ , but Carol's statement requires  $\neg a \wedge \neg b$ , i.e.  $a = \mathbf{F}$  — contradiction. Case  $a = \mathbf{F}$ : then  $b = \mathbf{T}$ ,  $c = \mathbf{F}$ . Check Carol: she lies, and indeed  $\neg a \wedge \neg b = \mathbf{F}$  — consistent. So Bob alone tells the truth. ■